

Differentially Private Preference Data Synthesis for Large Language Model Alignment

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Abstract

Preference alignment constitutes an essential post-training procedure for large language models (LLMs), calibrating model outputs to human-specified value criteria. Post-training on empirical human preference data, however, induces privacy risk: such datasets routinely encode sensitive user prompts alongside human comparative judgments. To mitigate this risk, we introduce **DPPrefSyn**, an algorithm that synthesizes differentially private (DP) preference data and thereby enables privacy-preserving preference alignment. DPPrefSyn is a principled framework predicated on the Bradley–Terry preference model and on the geometric structure induced by pairwise human preference comparisons. The procedure operates in two stages: it first estimates a latent preference model from the private data under formal differential privacy guarantees, and subsequently couples the estimated model with public prompts to synthesize high-fidelity preference pairs. To characterize heterogeneous human preferences within the private dataset, the method exploits the shared linear structure of per-cluster reward functions, and it applies DP Principal Component Analysis (DP-PCA) to reduce dimensionality and thereby improve estimation accuracy. Empirical evaluation across multiple benchmarks demonstrates that DPPrefSyn attains competitive alignment utility under stringent DP guarantees. These results establish synthetic preference data as a practical instrument for privacy-preserving preference alignment across a broad range of deployment settings. To the best of our knowledge, this constitutes the first method to generate DP synthetic preference data for LLM alignment. Our code will be released upon acceptance.

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1. Introduction

Preference alignment methods, including RLHF (Stiennon et al., 2020) and DPO (Rafailov et al., 2024), are the predominant mechanisms for aligning LLM behavior with human expectations. These methods operate on pairwise preference data: given a prompt, a human annotator compares two candidate responses and designates the one that better satisfies the prompt. The resulting preference labels supervise fine-tuning of the LLM through an objective that assigns higher likelihood to preferred responses than to dispreferred ones (Ziegler et al., 2019; Ouyang et al., 2022; Bai et al., 2022). The empirical efficacy and reliability of this paradigm have driven its adoption across diverse applications, including conversational assistants (Achiam et al., 2023), mathematical reasoning systems (Shao et al., 2024), and code generation models (Shen et al., 2023).

Aligning LLMs with human preference data nonetheless incurs substantial privacy risk, because such datasets typically comprise authentic user prompts together with human feedback. The prompts may expose personally identifiable information pertaining to health, identity, or other sensitive domains, while the feedback may reveal private beliefs, preferences, or behavioral patterns (Li et al., 2023; Yu et al., 2024). Prior work (Chowdhury et al., 2024; Yu et al., 2024; Wu et al., 2023a) has sought to bound these risks using the formal guarantees of differential privacy (DP) (Dwork et al., 2006b), with encouraging results. These approaches, however, privatize only a subset of the data—either the user prompts (Yu et al., 2024) or the preference labels (Chowdhury et al., 2024; Zhang et al., 2025), but not both. Such partial protection admits residual leakage, since an adversary may infer sensitive attributes from the unprotected prompts or preference annotations. Furthermore, these methods are typically instantiated for a specific post-training algorithm, such as RLHF (Wu et al., 2023a), and do not transfer to more recent methods such as DPO. Finally, they are constrained by the limited size of available private preference datasets and consequently attain only modest alignment quality, owing to the high acquisition cost and limited scalability of human preference annotation. These limitations motivate the central research question of this work:

Can superior preference alignment be attained while

simultaneously guaranteeing comprehensive privacy protection and preserving compatibility with emerging alignment methodologies?

This work answers the question in the affirmative. Our principal contributions are as follows:

- We introduce DPPrefSyn (Figure 1), an algorithm that synthesizes DP preference data for LLM alignment. DPPrefSyn is a principled framework predicated on the Bradley–Terry preference model and on the geometric structure induced by pairwise human preference comparisons. It comprises three design components: 1) *Modeling heterogeneous human preferences via a shared linear reward structure and clustering*. Under the Bradley–Terry model with a linear reward parameterization, the sign and magnitude of the inner product between the user preference vector and the embedding difference of a concatenated prompt–preferred and prompt–dispreferred pair govern the preference probability; this permits partitioning preference samples by their embedding differences and approximating the preferences within each partition by a shared reward function. 2) *Improving statistical efficiency via DP-PCA*. To counter the sample inefficiency and clustering instability inherent to high-dimensional embedding spaces, we apply DP-PCA to project embeddings onto a low-dimensional subspace that retains the dominant preference signal under DP. 3) *Conserving the privacy budget via public prompts*. We generate candidate responses from public prompts and select preference pairs using the privately estimated reward models, thereby allocating the entire privacy budget to modeling private preferences and improving privacy efficiency.
- We establish a rigorous privacy analysis of DPPrefSyn, proving that it satisfies (ϵ, δ) -DP. The analysis partitions the total budget as follows: a component ϵ_0 is allocated to DP-PCA (Amin et al., 2019) for dimensionality reduction, a component ϵ_1 to DP-KMeans (Su et al., 2016) for clustering, and the residual budget to training the per-cluster reward models via DP-SGD. We compose the per-iteration DP-SGD privacy loss using the Privacy Random Variable (PRV) accountant (Gopi et al., 2021), which yields a tight bound and certifies that DPPrefSyn satisfies (ϵ, δ) -DP. By the post-processing invariance of DP, the resulting synthetic dataset can be reused across arbitrary preference alignment methods and LLMs without additional privacy expenditure.
- We empirically evaluate DPPrefSyn on standard benchmarks spanning two task families: question answering, using OpenAssistant (Köpf et al., 2023) and Anthropic-HH (Bai et al., 2022), and summarization, using TL;DR (Stiennon et al., 2020). Across these tasks, the synthetic preference data produced by DPPrefSyn attains competitive utility while satisfying stringent privacy guarantees.

As a concrete instance, at $\epsilon = 2$, fine-tuning Pythia-2.8B with DPO on our synthetic data attains a GPT-4o win rate of 56.48% on Anthropic-HH, substantially exceeding the 37.02% obtained by fine-tuning on the real data. These results demonstrate that DP synthetic preference data suffices for effective LLM alignment under strong privacy constraints. Moreover, DPPrefSyn is agnostic to the downstream preference optimization objective—it applies to DPO, RLHF, and other methods—and remains effective in the high-privacy regime, thereby overcoming the method-specific restrictions of prior privacy-preserving post-training approaches.

2. Related Work

Differentially Private Alignment of LLMs. Wu et al. (2023a) propose a DP framework for reinforcement-learning-based alignment that adapts the PPO algorithm to the DP setting; the construction is, however, *specific to PPO*, whereas the preference alignment literature continues to introduce new objectives. Chowdhury et al. (2024) address reward estimation from preference feedback, invoking label-DP (Ghazi et al., 2021) to protect annotator privacy. Zhang et al. (2025) propose AUP-RLHF, a user-level label-DP framework for RLHF. Although both methods effectively protect preference labels, neither *protects the prompts or responses*. Moreover, both are *constrained by the limited quantity of private preference data* and therefore attain only modest alignment quality, since human preference annotations are costly to acquire. Yu et al. (2024) instead protect sensitive user *instructions*, synthesizing DP *instructions* that substitute for the real ones during annotation and fine-tuning.

Beyond preference alignment, a separate body of work addresses private fine-tuning of LLMs on general labeled data rather than preference comparisons (Yu et al., 2021; Li et al., 2021; Chen et al., 2024; Tang et al., 2024; Zhang et al., 2023; He et al., 2022), while a further line protects the privacy of in-context prompts for in-context learning (Duan et al., 2023; Tang et al., 2023; Gao et al., 2024; Hong et al., 2023; Wu et al., 2023b).

Differentially Private Synthetic Text Generation. Our work belongs to the broader area of DP synthetic text generation, within which two representative methodologies have emerged. The first fine-tunes a language model on the private data under DP and then samples synthetic text from the fine-tuned model (Yue et al., 2023; Mattern et al., 2022; Mireshghallah et al., 2023; Carranza et al., 2023; Yu et al., 2024; Wang et al., 2024; Ochs & Habernal, 2024; Tan et al., 2025; Carranza et al., 2024). For instance, Yu et al. (2024) privately fine-tune the LLaMA model on private instruction data and subsequently sample synthetic text from it for supervised instruction tuning.

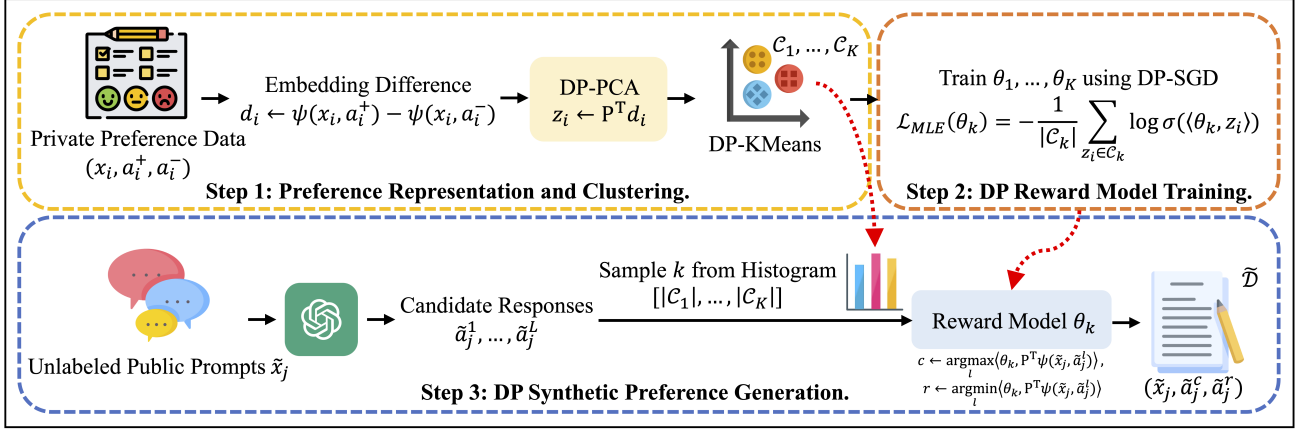


Figure 1. Overview of DPPrefSyn. The algorithm synthesizes DP preference data via three stages: 1) it represents each preference sample as an embedding difference and clusters these differences using DP-PCA followed by DP-KMeans; 2) it trains a DP reward model on each cluster via DP-SGD; and 3) it generates synthetic preference pairs from public prompts, selected according to the DP-protected distribution over per-cluster reward functions.

Because DP fine-tuning is computationally expensive and frequently infeasible—in particular for closed-weight language models—a second methodology generates DP synthetic data by querying LLM APIs (Lin et al., 2023; Xie et al., 2024; Wu et al., 2024; Hou et al., 2024). These methods sample candidate outputs through API access and then select those that are similar to the private data under a privacy-preserving selection rule. For example, Xie et al. (2024) propose Aug-PE, which iteratively refines LLM-generated samples according to their embedding-space similarity to the private data. *Our work follows this second methodology and specializes it to preference alignment of LLMs.*

Diversity in Human Preferences. A growing body of work documents that human preferences are intrinsically heterogeneous (Denton et al., 2021; Aroyo et al., 2023a,b; Chakraborty et al., 2024). The principal sources of this heterogeneity include sociodemographic factors (e.g., race, gender, age), individual beliefs and biases, differing levels of domain expertise, and the inherent ambiguity of natural language (Sandri et al., 2023; Vogels, 2021). Motivated by these observations, recent studies personalize preference alignment to better reflect the values of distinct user populations (Lee et al., 2024b; Poddar et al., 2024; Singh et al., 2025). For instance, Singh et al. (2025) formulate a meta-learning framework in which an LLM adapts to an individual user from a small set of preference examples. In this work, we synthesize DP preference data that *explicitly preserves the underlying diversity of human feedback*, which yields improved alignment relative to methods that disregard this diversity (Section 5.3).

3. Problem Definition

Preference Alignment of LLMs. We consider prefer-

ence alignment of an LLM from pairwise human feedback, formalized through a preference dataset $\mathcal{D} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$ of n triples. Each triple consists of a prompt x_i together with two candidate responses a_i^+, a_i^- , where the annotator designates a_i^+ as preferred over a_i^- .

Following the standard formulation adopted in reinforcement learning from human feedback (RLHF) and direct preference optimization (DPO) (Ouyang et al., 2022; Zhu et al., 2023; Rafailov et al., 2024; Liu et al., 2023), we model the annotator’s preference as a Bradley–Terry choice model (Bradley & Terry, 1952):

$$\begin{aligned} \mathbb{P}[a_i^+ \succ a_i^- \mid x_i] &= \frac{\exp(r^*(x_i, a_i^+))}{\exp(r^*(x_i, a_i^+)) + \exp(r^*(x_i, a_i^-))} \\ &= \sigma(r^*(x_i, a_i^+) - r^*(x_i, a_i^-)), \end{aligned}$$

where the relation $a_i^+ \succ a_i^-$ denotes that a_i^+ is preferred to a_i^- , $r^*(x, a)$ denotes the latent reward function, and $\sigma(z) = 1/(1 + e^{-z})$ denotes the logistic sigmoid.

This model underlies both RLHF, which estimates an explicit reward model from the preference data, and DPO, which induces an implicit reward equal to the log-ratio between the learned policy and a fixed reference policy.

Differentially Private Preference Data Synthesis. Our objective is to protect the private preference dataset $\mathcal{D}_{\text{priv}} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$ against an adversary that seeks to access or infer private information about individual prompts, responses, or preference labels from the outputs of the fine-tuned LLM. To this end, we synthesize a DP preference dataset $\tilde{\mathcal{D}} = \{(\tilde{x}_j, \tilde{a}_j^+, \tilde{a}_j^-)\}_{j=1}^m$ of m triples that retains the salient preference information contained in $\mathcal{D}_{\text{priv}}$. By the post-processing invariance of DP, $\tilde{\mathcal{D}}$ can be reused across arbitrary preference alignment methods and LLMs without incurring additional privacy cost.

We recall the formal definition of (ϵ, δ) -differential privacy.

Definition 3.1 ((ϵ, δ) -Differential Privacy (Dwork et al., 2006a)). A randomized algorithm $\mathcal{A}$ is (ϵ, δ) -differentially private if for any two neighboring inputs $\mathcal{D}$ and $\mathcal{D}'$ that differ by a single entry and any set $\mathcal{S}$ of possible outputs: $\mathbb{P}[\mathcal{A}(\mathcal{D}) \in \mathcal{S}] \leq e^\epsilon \mathbb{P}[\mathcal{A}(\mathcal{D}') \in \mathcal{S}] + \delta$.

Diverse User Preferences and Linear Reward Structure.

The preferences represented in the private dataset are generally heterogeneous (Chakraborty et al., 2024; Bakker et al., 2022; Kovač et al., 2023; Jang et al., 2023; Rame et al., 2023; Ji et al., 2023). Annotators weight distinct attributes of a response—for example, factual accuracy, politeness, creativity, or clarity—differently, which produces diverse and at times mutually conflicting supervision signals. Faithfully synthesizing $\bar{\mathcal{D}}$ therefore requires capturing this heterogeneity in $\mathcal{D}_{\text{priv}}$.

To this end, we impose a linear parameterization on the latent reward function (Saha et al., 2023; Kong & Yang, 2022; Zhu et al., 2023). Concretely, we adopt the linear reward model

$$r_\theta(x, a) = \langle \theta, \phi(x, a) \rangle,$$

in which $\phi(x, a) \in \mathbb{R}^d$ denotes the *shared* feature map and $\theta \in \mathbb{R}^d$ denotes a *user-dependent* weight vector.

This parameterization trades off *expressiveness* against *statistical and computational tractability* in modeling diverse preferences. Although users differ in their preferences, their reward functions share a common set of semantic features encoded by the learned embedding $\phi(x, a)$ and differ principally in the weighting θ assigned to those features. The linear form isolates this shared structure while permitting variation across users or preference clusters, which enables tractable estimation of preference heterogeneity under limited data and stringent privacy constraints. As the empirical study in Section B.6 confirms, the linear reward model captures the preference signal effectively and performs competitively with a fully fine-tuned GPT-2 reward model despite its comparative simplicity.

4. Proposed Method

4.1. Key Components of the Algorithm Design

Before presenting the full algorithm, we describe the components that address the principal challenges in synthesizing differentially private preference data.

Exploiting the Linear Reward Structure for Preference Clustering. Under the linear reward model, the Bradley–Terry probability factors as

$$\mathbb{P}[a_i^+ \succ a_i^- \mid x_i] = \sigma(\langle \theta, \phi(x_i, a_i^+) - \phi(x_i, a_i^-) \rangle).$$

Consequently, for a triple (x_i, a_i^+, a_i^-) , the observed preference $a_i^+ \succ a_i^-$ implies $\langle \theta, \phi(x_i, a_i^+) - \phi(x_i, a_i^-) \rangle \gg 0$; equivalently, the weight vector θ is strongly aligned with the feature difference $\phi(x_i, a_i^+) - \phi(x_i, a_i^-)$. It follows that, for annotators sharing similar preferences, the feature difference vectors $\phi(x_i, a_i^+) - \phi(x_i, a_i^-)$ over their rated pairs concentrate along a common direction in the embedding space. This observation motivates clustering the preference samples by their feature difference vectors $\phi(x_i, a_i^+) - \phi(x_i, a_i^-)$, so that the preferences within each cluster admit an accurate approximation by a single shared reward function parameterized by a common weight vector θ . Section B.9 demonstrates that the clusters obtained by this procedure are semantically meaningful and interpretable as distinct user preference styles.

Applying DP-PCA to Obtain Compact Features. A natural instantiation of the feature map $\phi(x, a)$ is the embedding of the concatenated (prompt, response) pair, since such embeddings encode the semantic similarities and distinctions that underlie human preference judgments. These embeddings are, however, typically high-dimensional (e.g., $d = 1024$) in order to represent rich semantic and stylistic content. Estimation in a space of this dimensionality demands a sample size that exceeds the number of samples available within an individual cluster. To resolve this mismatch, we apply differentially private PCA (DP-PCA) (Amin et al., 2019; Liu et al., 2022) to reduce the embedding dimensionality and produce compact representations for both clustering and reward-model estimation. This projection improves sample efficiency and clustering stability while retaining the preference signal required for effective alignment.

Leveraging Public Prompts to Conserve the DP Budget. Jointly synthesizing prompts, preferred responses, and dispreferred responses would rapidly deplete the privacy budget and degrade utility. We instead draw prompts from a public corpus, which eliminates the need to expend privacy budget on prompt synthesis and permits allocating the entire budget to modeling preferences over responses. For each public prompt, we sample multiple candidate responses from an LLM and construct preference pairs using the distribution over reward functions estimated from the private data under DP. Although the public prompt distribution may differ from the private one, our experiments show that this distribution shift does not compromise the synthetic data: it preserves the statistical preference distribution of the private dataset and remains effective for downstream alignment.

4.2. DPPrefSyn Algorithm

We now specify the DPPrefSyn algorithm (Algorithm 1), which maps the private dataset $\mathcal{D}_{\text{priv}}$ to a DP synthetic preference dataset. The procedure comprises three steps:

Step 1: Preference Representation and Clustering. We encode each prompt–response pair in $\mathcal{D}_{\text{priv}}$ using a public sentence embedding model ψ . For each triple (x_i, a_i^+, a_i^-) , we concatenate the prompt with each response to form the two texts $[x_i; a_i^+]$ and $[x_i; a_i^-]$, embed both via ψ , and take their difference $d_i \leftarrow \psi(x_i, a_i^+) - \psi(x_i, a_i^-)$ (Line 3).

To reduce the dimensionality of these representations under DP, we apply DP-PCA (Amin et al., 2019) to $\{d_i\}_{i=1}^n$, which returns a projection matrix $\mathbf{P} \in \mathbb{R}^{d \times p}$ satisfying ε_0 -DP (Line 5). DP-PCA (Amin et al., 2019) approximates the eigendecomposition of the data covariance matrix by privately estimating its leading eigenvalues and eigenvectors as two separate DP subroutines. Each difference vector is then projected onto the estimated subspace, $z_i \leftarrow \mathbf{P}^\top d_i$ (Line 6). We subsequently apply DP-KMeans (Su et al., 2016) to $\{z_i\}_{i=1}^n$ to partition the projected samples into K clusters $\mathcal{C}_1, \dots, \mathcal{C}_K$ under ε_1 -DP (Line 7); DP-KMeans (Su et al., 2016) is a differentially private instantiation of Lloyd’s algorithm.

Step 2: DP Reward Model Training. For each cluster $\mathcal{C}_k$, we estimate the shared weight vector $\theta_k \in \mathbb{R}^p$ that characterizes the cluster’s preference pattern by minimizing the negative log-likelihood of the Bradley–Terry model (Line 9):

$$\mathcal{L}_{\text{MLE}}(\theta_k) = -\frac{1}{|\mathcal{C}_k|} \sum_{z_i \in \mathcal{C}_k} \log \sigma(\langle \theta_k, z_i \rangle), \quad (1)$$

where z_i denotes the DP-PCA-projected embedding difference from Step 1. We optimize each objective with DP-SGD (Abadi et al., 2016), the differentially private variant of stochastic gradient descent, using noise multiplier σ_2 .

Step 3: DP Synthetic Preference Generation. We form the cluster-size histogram $\mathbf{h} = [|\mathcal{C}_1|, \dots, |\mathcal{C}_K|]$ (Line 11) and normalize it to a categorical distribution $\mathbf{p}$ over the K clusters, $\mathbf{p} \leftarrow \mathbf{h}/|\mathcal{D}_{\text{priv}}|$ (Line 12).

For each public prompt $\tilde{x}_j$, we sample L candidate responses $\tilde{a}_j^1, \dots, \tilde{a}_j^L$ from an LLM at elevated temperature to promote diversity (Line 14). We then draw a cluster index $k \sim \mathbf{p}$ from the DP histogram (Line 15) and score the candidates under the reward model θ_k : for each $l = 1, \dots, L$, we compute the embedding $\psi(\tilde{x}_j, \tilde{a}_j^l)$, project it onto the DP-PCA subspace via $\mathbf{P}$, and evaluate the reward $\langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$. We designate the highest- and lowest-scoring candidates as the preferred and dispreferred responses, respectively: $c \leftarrow \arg \max_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ and $r \leftarrow \arg \min_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ (Lines 16–17). To ensure a sufficient preference margin, we discard the triple whenever the reward gap falls below a threshold (e.g., 0.5); otherwise we append $(\tilde{x}_j, \tilde{a}_j^c, \tilde{a}_j^r)$ to $\tilde{\mathcal{D}}$. Iterating over all public prompts yields the DP synthetic preference dataset $\tilde{\mathcal{D}}$.

Algorithm 1 DPPrefSyn

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1: Input: Private dataset  $\mathcal{D}_{\text{priv}} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$ , public prompts  $\{\tilde{x}_j\}_{j=1}^m$ , embedding model  $\psi$ , number of clusters  $K$ , an LLM  $\text{LLM}(\cdot)$ , DP parameters  $\varepsilon_0, \varepsilon_1, \sigma_2$ .
2: for each  $(x_i, a_i^+, a_i^-)$  in  $\mathcal{D}_{\text{priv}}$  do
3:    $d_i \leftarrow \psi(x_i, a_i^+) - \psi(x_i, a_i^-)$ , where  $d_i \in \mathbb{R}^d$ 
4: end for
5:  $\mathbf{P} \leftarrow \text{DP-PCA}(\{d_i\}_{i=1}^n, \varepsilon_0)$ , where  $\mathbf{P} \in \mathbb{R}^{d \times p}$ 
6:  $\{z_i\}_{i=1}^n \leftarrow \{\mathbf{P}^\top d_i\}_{i=1}^n$ 
7:  $\mathcal{C}_1, \dots, \mathcal{C}_K \leftarrow \text{DP-KMeans}(\{z_i\}_{i=1}^n, K, \varepsilon_1)$ 
8: for each cluster  $\mathcal{C}_k$  do
9:   Train a linear reward model  $\theta_k$  on  $\mathcal{C}_k$  using DP-SGD with noise multiplier  $\sigma_2$  (Equation (1))
10: end for
11:  $\mathbf{h} \leftarrow [|\mathcal{C}_1|, \dots, |\mathcal{C}_K|]$ 
12:  $\mathbf{p} \leftarrow \mathbf{h}/|\mathcal{D}_{\text{priv}}|$ 
13: for each public prompt  $\tilde{x}_j$  do
14:   Generate responses  $\tilde{a}_j^1, \dots, \tilde{a}_j^L \leftarrow \text{LLM}(\tilde{x}_j)$ 
15:   Sample cluster index  $k \sim \mathbf{p}$ 
16:    $c \leftarrow \arg \max_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ 
17:    $r \leftarrow \arg \min_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ 
18:   Add  $(\tilde{x}_j, \tilde{a}_j^c, \tilde{a}_j^r)$  to  $\tilde{\mathcal{D}}$ 
19: end for
20: return  $\tilde{\mathcal{D}}$ .
    
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Privacy Analysis of DPPrefSyn. We partition the total budget by assigning ε_0 to DP-PCA (Line 5) and ε_1 to DP-KMeans (Line 7), and allocating the residual budget to per-cluster reward-model training via DP-SGD with noise multiplier σ_2 (Line 9). For the reward-model training step we invoke parallel composition: because the clusters are disjoint, modifying a single training point perturbs exactly one cluster, so the aggregate privacy cost of this step is bounded by the cost incurred on the smallest cluster. We enforce a known lower bound on the cluster size by discarding clusters with too few samples. We compose the per-step DP-SGD privacy costs with the Privacy Random Variable (PRV) accountant (Gopi et al., 2021), which certifies that this step satisfies $(\varepsilon - \varepsilon_0 - \varepsilon_1, \delta)$ -DP. Every remaining step operates solely on public data or is a post-processing of DP outputs, and therefore incurs no additional privacy cost. The complete proof and the privacy-accounting code appear in Section A.

Robustness to Prompt Distribution Shift. DPPrefSyn remains effective under distribution shift between the private and public prompt distributions. The key observation is that preference heterogeneity is encoded in the reward-model weight vector θ_k rather than in the feature map $\phi(x, a)$ of the prompt–response pair: a given user may issue widely varying prompts while their underlying preference remains fixed. Consequently, provided that the distribution over θ_k in the synthetic data matches that in the private data,

the synthesized preferences continue to reflect the private preference patterns. We enforce this match by drawing, for each public prompt, a cluster index k from the DP histogram (Line 15) and selecting the response pair under θ_k . Thus, DPPrefSyn preserves the private preference structure even when the private and public prompt distributions differ.

5. Experiments

Datasets. We evaluate on two task families, question answering and summarization. For question answering, we use OpenAssistant (Köpf et al., 2023), comprising assistant-style conversations, and Anthropic-HH (Bai et al., 2022), comprising human preference comparisons over helpfulness and harmlessness. For summarization, we use TL;DR (Stienon et al., 2020), comprising human preference annotations over pairs of summaries. As the public prompt corpus, we use Alpaca (Taori et al., 2023) for the OpenAssistant QA task, SafeRLHF (Ji et al., 2024) for the Anthropic-HH QA task, and XSum (Narayan et al., 2018) for the TL;DR summarization task. Further details are provided in Section B.2.

Configuration of Algorithm 1. Unless otherwise stated, we instantiate the embedding model ψ as BAAI/bge-large-en-v1.5 (Xiao et al., 2023), and set the DP-PCA projected dimension to $p = 20$ and the number of DP-KMeans clusters to $K = 5$. DP-SGD uses a learning rate of 0.1, a batch size of 4, 4 epochs, and a gradient clipping norm of 1.0. Candidate responses are sampled from the instruction-tuned LLaMA-7B-chat model (Touvron et al., 2023) at temperature 0.9, with $L = 5$ candidates per prompt. We report results across total privacy budgets $\varepsilon \in \{0.5, 1, 2, 4, 8\}$ and fix $\delta = 1/|\mathcal{D}_{\text{priv}}|$. *The effect of the generator model is examined in Table 3, and the effects of the embedding model and remaining hyperparameters in Section 5.3.*

Evaluation. For downstream preference alignment, we fine-tune Pythia-2.8B (Biderman et al., 2023). We first perform supervised fine-tuning (SFT), taking the preferred response in each pair as the training target, and then apply DPO (Rafailov et al., 2024) to further fine-tune the SFT model on the preference pairs. Following Rafailov et al. (2024), we quantify utility as the win rate of the model’s responses against the preferred test-set responses, adjudicated by GPT-4o. *Additional downstream models and alternative preference alignment methods are examined as ablations in Section 5.2.*

Baselines. Our principal baseline, denoted DP-FT, fine-tunes the downstream model directly on the real data via DP-SGD (Abadi et al., 2016). DPPrefSyn is strictly more reusable than DP-FT: it emits a DP synthetic dataset that trains arbitrary models under arbitrary alignment algorithms at no additional privacy cost, whereas DP-FT expends its

entire budget producing a single privatized model. We additionally include a fully private reference point, $\varepsilon = 0$, corresponding to the base model evaluated without any preference fine-tuning.

Complete setups, hyperparameters, metrics, and baseline configurations are deferred to Section B.3.

5.1. Understanding the Performance of DPPrefSyn

In this section, we characterize the privacy–utility trade-off of DPPrefSyn through two research questions. The main results are reported in Table 1.

RQ1: Does DPPrefSyn dominate fine-tuning on real private data? Across all tasks and every finite budget $\varepsilon \in \{0.5, 1, 2, 4, 8\}$, DPPrefSyn attains a strictly superior privacy–utility trade-off relative to DP-FT, and it exceeds the utility of unconstrained fine-tuning on real private data (DP-FT at $\varepsilon = \infty$) while retaining privacy. On the OpenAssistant QA task at $\varepsilon = 2$, DPPrefSyn attains 9.10% after SFT and 11.04% after DPO, versus 4.38% and 6.18% for DP-FT. Notably, DPPrefSyn also exceeds the non-private DP-FT reference ($\varepsilon = \infty$), which attains 4.75% after SFT and 8.20% after DPO, while continuing to satisfy the privacy guarantee. The same pattern holds on Anthropic-HH at $\varepsilon = 2$: DPPrefSyn attains a 55.21% win rate after SFT and 56.48% after DPO, against 31.53% and 37.02% for DP-FT, and again surpasses non-private fine-tuning on real data (DP-FT at $\varepsilon = \infty$), which attains 31.98% after SFT and 38.72% after DPO.

These results reveal two **principal advantages** of the DP synthetic data. **First**, DPPrefSyn yields a substantial gain after SFT alone relative to SFT on private data. On the TL;DR summarization task at $\varepsilon = 2$, for example, DPPrefSyn attains a 62.49% win rate after SFT, far exceeding the 22.57% obtained by SFT on private data. This gain is attributable to the high-quality candidate responses sampled from a strong generator LLM, which furnish more informative supervision during SFT. The advantage persists under limited public data: on the OpenAssistant QA task at $\varepsilon = 4$ with only 5K public prompts, DPPrefSyn attains 6.80% after SFT, against 4.46% for SFT on the roughly 14K private samples (Table 4), evidencing robustness to data scarcity. **Second**, applying DPO to the SFT model on the synthetic pairs produces a further gain, indicating that the synthetic preference pairs encode meaningful, human-aligned distinctions; this is a direct consequence of the private reward functions, which capture diverse user preferences under DP. The incremental gain from DPO is smaller than in the private-data setting, plausibly because the SFT model trained on synthetic data is already near-optimal and leaves less residual headroom. Nonetheless, over the full SFT+DPO pipeline, the DP synthetic data yields a substantially larger improvement than the real private data.

Table 1. GPT-4o win rates on the test sets for Pythia-2.8B fine-tuned by SFT and by SFT+DPO. The column $\varepsilon = 0$ reports the base (fully private, non-fine-tuned) LLM. At every finite budget $\varepsilon \in \{0.5, 1, 2, 4, 8\}$, DPPrefSyn attains higher win rates than DP-FT, and it even exceeds the utility of non-private fine-tuning (DP-FT at $\varepsilon = \infty$). Entries report the mean and standard deviation over 5 random seeds.

Task	$\varepsilon = 0$	Method	Data Type	Pref. Alignment	$\varepsilon = 0.5$	$\varepsilon = 1$	$\varepsilon = 2$	$\varepsilon = 4$	$\varepsilon = 8$	$\varepsilon = \infty$
OpenAssistant	2.11	DP-FT	Original	SFT	3.96 _{0.20}	4.29 _{0.21}	4.38 _{0.23}	4.46 _{0.18}	4.49 _{0.30}	4.75 _{0.27}
				SFT+DPO	5.65 _{0.32}	6.04 _{0.58}	6.18 _{0.83}	6.12 _{0.63}	6.32 _{0.71}	8.20 _{0.62}
		DPPrefSyn	Synthetic	SFT	8.96 _{0.39}	9.07 _{0.32}	9.10 _{0.51}	9.33 _{0.56}	9.63 _{0.38}	9.80 _{0.58}
				SFT+DPO	9.86_{0.59}	10.31_{0.54}	11.04_{0.92}	11.85_{0.99}	12.05_{0.82}	12.36_{0.55}
Anthropic-HH	12.14	DP-FT	Original	SFT	29.77 _{0.76}	31.10 _{0.72}	31.53 _{0.77}	31.29 _{0.65}	31.37 _{0.67}	31.98 _{0.56}
				SFT+DPO	35.00 _{0.49}	36.27 _{0.63}	37.02 _{0.72}	36.74 _{0.90}	36.94 _{1.06}	38.72 _{0.56}
		DPPrefSyn	Synthetic	SFT	54.90 _{0.66}	55.19 _{0.91}	55.21 _{0.82}	54.95 _{1.10}	55.63 _{0.86}	55.95 _{0.70}
				SFT+DPO	55.08_{1.04}	55.96_{0.80}	56.48_{0.61}	56.51_{1.93}	56.86_{1.17}	57.53_{0.93}
TL;DR	11.64	DP-FT	Original	SFT	21.19 _{1.12}	22.41 _{1.09}	22.57 _{0.65}	22.45 _{0.90}	23.04 _{0.79}	23.82 _{1.26}
				SFT+DPO	59.69 _{0.82}	62.21 _{1.12}	62.02 _{1.67}	63.00 _{0.73}	63.06 _{0.84}	64.52 _{1.79}
		DPPrefSyn	Synthetic	SFT	61.92 _{1.06}	62.14 _{1.61}	62.49 _{1.48}	62.72 _{1.66}	62.29 _{1.14}	62.55 _{1.23}
				SFT+DPO	67.50_{0.79}	68.07_{1.95}	68.54_{2.76}	68.56_{1.59}	71.01_{2.17}	72.20_{1.08}

Table 2. On OpenAssistant, DPPrefSyn attains higher win rates than fine-tuning on private data under each of the SFT, DPO, and PPO objectives, while satisfying strong privacy guarantees. Entries report the mean and standard deviation over 5 random seeds.

Pref. Alignment	$\varepsilon = 0$	DPPrefSyn ($\varepsilon = 4$)	DP-FT ($\varepsilon = \infty$)
SFT		9.33_{0.56}	4.75 _{0.27}
SFT+DPO	2.11	11.85_{0.99}	8.20 _{0.62}
SFT+PPO		10.11_{0.61}	7.08 _{0.78}

RQ2: How does DPPrefSyn behave as a function of the privacy budget ε ? **First**, Table 1 shows that DPPrefSyn dominates DP-FT across the full range $\varepsilon \in \{0.5, 1, 2, 4, 8, \infty\}$, and that its utility scales gracefully with ε . **Second**, DPPrefSyn is in some regimes markedly more robust to DP noise than DP-FT. On OpenAssistant with Llama-3-8B as the downstream model, the DPO win rate of DP-FT degrades from 59.13% to 45.79% as ε decreases from ∞ to 4, whereas DPPrefSyn degrades only from 61.52% to 60.25% (Table 3, Section 5.2). We attribute this contrast to the locus of noise injection: DP-FT perturbs the model parameters directly through DP-SGD, whereas DPPrefSyn injects noise only during data synthesis, which better preserves downstream utility.

5.2. Understanding the Properties of DPPrefSyn

In this section, we characterize properties of DPPrefSyn: its generalization across preference alignment methods and downstream LLMs, its behavior as a function of public-data scale, and its resilience to empirical privacy attacks.

RQ3: Does DPPrefSyn generalize across preference alignment methods? **DPPrefSyn is compatible with a range of preference alignment objectives and remains effective under strong DP constraints.** Table 2 evaluates DPPrefSyn under SFT, DPO, and PPO. For PPO, we train the K per-cluster reward models and define the scalar reward as the mean of their outputs, then optimize the policy against this averaged reward. Across all three objectives, DPPref-

Table 3. On OpenAssistant, DPPrefSyn attains higher post-DPO win rates than fine-tuning on private data across distinct downstream LLMs, while satisfying strong privacy guarantees. Utility further increases with a stronger generator LLM.

Downstream LLM	$\varepsilon = 0$	Method	Generator LLM	$\varepsilon = 4$	$\varepsilon = \infty$
Qwen-3-4B	48.17	DP-FT	-	54.92	59.41
			-		
		DPPrefSyn	Qwen-3-4B-Instruct	86.52	87.08
			Llama-3-8B-Instruct	51.69	52.11
Gemma-2-2B	7.30	DP-FT	-	30.06	33.85
			-		
		DPPrefSyn	Qwen-3-4B-Instruct	36.24	37.08
			Llama-3-8B-Instruct	36.01	36.66
Llama-3-8B	16.15	DP-FT	-	45.79	59.13
			-		
		DPPrefSyn	Qwen-3-4B-Instruct	60.25	61.52
			Llama-3-8B-Instruct	42.28	44.10

Table 4. DPPrefSyn attains higher win rates than fine-tuning on private data across public-data sizes m . Entries report the mean and standard deviation over 5 random seeds.

Data Type (Size)	DPPrefSyn ($\varepsilon = 4$)			DP-FT ($\varepsilon = 4$)	
	Synthetic ($m = 5,000$)	Synthetic ($m = 10,000$)	Synthetic ($m = 52,000$)	Original (14,167)	Original (14,167)
SFT	6.80 _{0.66}	7.30 _{0.48}	9.33 _{0.56}	4.46 _{0.18}	4.75 _{0.27}
SFT+DPO	9.44 _{0.45}	9.55 _{0.92}	11.85 _{0.99}	6.12 _{0.63}	8.20 _{0.62}

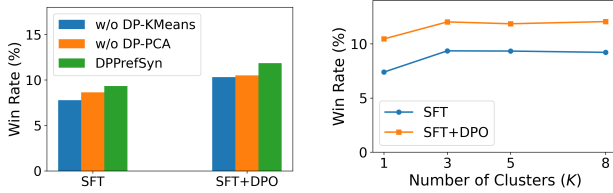
Syn attains higher utility than fine-tuning on private data while preserving strong privacy guarantees.

RQ4: Do stronger downstream LLMs benefit from DPPrefSyn’s synthetic data? **DPPrefSyn is effective across a broad range of downstream LLMs.** Table 3 evaluates three downstream models—Qwen-3-4B, Gemma-2-2B, and Llama-3-8B—with Llama-3-8B-Instruct and Qwen3-4B-Instruct-2507 as generators. Post-DPO, DPPrefSyn surpasses fine-tuning on real data across all model families while preserving privacy. With Qwen3-4B-Instruct-2507 as the generator, for instance, DPPrefSyn attains an 86.52% win rate on Qwen-3-4B at $\varepsilon = 4$, substantially exceeding non-private fine-tuning on real data (59.41%). Utility further increases with generator strength: Qwen3-4B-Instruct-2507 consistently outperforms Llama-3-8B-Instruct.

RQ5: How does DPPrefSyn scale with the amount of

Table 5. On OpenAssistant, DPPrefSyn yields lower MIA AUC (Feng et al., 2025)—near the 50% random-guessing baseline—than DP-FT. Entries report the mean and standard deviation over 5 random seeds.

Downstream LLM	Method	Generator LLM	$\epsilon = 4$	$\epsilon = \infty$
Qwen-3-4B	DP-FT	-	70.15 _{2.09}	73.52 _{2.30}
	DPPrefSyn	Qwen-3-4B-Instruct	50.99 _{1.10}	50.87 _{0.84}
		Llama-3-8B-Instruct	50.64_{1.29}	49.98_{1.19}
Gemma-2-2B	DP-FT	-	62.88 _{1.78}	65.39 _{2.59}
	DPPrefSyn	Qwen-3-4B-Instruct	50.51_{1.38}	50.43_{0.91}
		Llama-3-8B-Instruct	50.77 _{0.83}	51.10 _{0.54}
Llama-3-8B	DP-FT	-	61.95 _{2.05}	70.13 _{1.24}
	DPPrefSyn	Qwen-3-4B-Instruct	50.53 _{0.62}	50.96 _{0.95}
		Llama-3-8B-Instruct	50.45_{0.87}	50.67_{0.96}



(a) Ablating DP-KMeans or DP-PCA. (b) Varying the cluster count K .

public data? Table 4 quantifies the effect of the public-data size m . The private dataset (OpenAssistant) comprises 14K preference annotations and the full public corpus (Alpaca) comprises 52K prompts; we synthesize preference data from uniformly sampled public subsets of sizes $m \in \{5K, 10K, 52K\}$. Even at $m = 5K$, DPPrefSyn attains 6.80% after SFT and 9.44% after DPO at $\epsilon = 4$, exceeding the DP-FT baseline’s 4.46% and 6.12%. Thus **DPPrefSyn remains effective under scarce public data**, a consequence of the high-quality LLM-generated responses and the private reward functions that faithfully capture user preferences. Moreover, **the margin over the baseline widens as m grows**, indicating that DPPrefSyn effectively converts abundant public data into alignment utility and thereby further mitigates the scarcity of private preference data.

RQ6: Does DPPrefSyn mitigate empirical privacy attacks? Table 5 reports the AUC of membership inference attacks (MIAs) (Shokri et al., 2017) mounted against the DPO fine-tuned downstream models on OpenAssistant. We adopt the attack of Feng et al. (2025), which instantiates a MIA specific to preference data for LLM alignment; the attacker must decide whether a given private triple (x, a^+, a^-) participated in preference alignment. DP-FT fails to adequately suppress this attack, whereas **DPPrefSyn consistently yields AUC near the 50% random-guessing baseline**. This indicates greater resilience to empirical privacy attacks, plausibly because fine-tuning on synthetic rather than real data structurally limits overfitting to the private samples. Full details appear in Section B.4.

5.3. Validating the Design of DPPrefSyn

We validate the design of DPPrefSyn through an ablation of its algorithmic components.

RQ7: What is the contribution of modeling preference diversity? We isolate this contribution by ablating DP-KMeans and DP-PCA and by varying the cluster count K , on OpenAssistant at $\epsilon = 4$. **(i) Ablating DP-KMeans or DP-PCA.** Figure 2a shows that removing either component degrades performance. Removing DP-KMeans collapses all samples into a single cluster; removing DP-PCA yields ineffective clustering and sample-inefficient estimation in the high-dimensional embedding space. Retaining both components and using multiple clusters strictly improves over the single-cluster configuration. **(ii) Varying the cluster count K .** Figure 2b evaluates $K \in \{1, 3, 5, 8\}$. Utility improves from $K = 1$ to $K = 3$ and then plateaus at $K \in \{5, 8\}$, indicating that DPPrefSyn is insensitive to K and that moderate values (e.g., $K \in \{3, 5\}$) balance preference diversity against per-cluster data efficiency. Collectively, these results confirm the benefit of representing preference diversity through multiple clusters.

RQ8: How sensitive is DPPrefSyn to the embedding model and hyperparameters? We ablate the embedding model ψ , the DP-PCA dimension p , the number of candidates per prompt L , the filtering threshold, and the privacy-budget allocation in Section B.5. Across the sentence encoders bge-large-en-v1.5 (Xiao et al., 2023), all-mpnet-base-v2 (Song et al., 2020), and all-distilroberta-v1 (Sanh et al., 2019), DPPrefSyn remains uniformly effective. The dimension $p = 20$ maximizes post-DPO utility; $L = 5$ serves as a robust default; a moderate filtering threshold of 0.5 marginally exceeds no filtering, whereas an aggressive threshold of 1 discards too many pairs and degrades performance. Finally, DPPrefSyn is robust to reasonable perturbations of the budget split (ϵ_0, ϵ_1) .

RQ9: Are DPPrefSyn’s gains attributable merely to higher-quality public prompts or stronger response generators? To control for these confounders, we run ablations in which all methods share an identical prompt set or an identical response generator. Under both controls, DPPrefSyn continues to outperform DP-FT (Appendix B.7), establishing that **its gains are not explained by prompt quality or generator strength alone**.

6. Conclusion

We introduced DPPrefSyn, an algorithm that synthesizes DP preference data from a private dataset. By representing preference diversity through clustering of embedding differences, improving statistical efficiency via DP-PCA, and conserving the privacy budget through public prompts,

DPPrefSyn enables effective and private preference-data synthesis for LLM alignment. Our experiments demonstrate that DPPrefSyn satisfies formal privacy guarantees while exceeding the utility of fine-tuning on real data.

7. Limitations

DPPrefSyn presupposes LLM API access for candidate-response generation, which may incur additional API cost. The present study is confined to English text-based preference data; extending DPPrefSyn to multilingual and multi-modal settings remains open. Finally, computational constraints restrict our experiments to models of at most 8B parameters.

Impact statement

This work enables preference alignment of large language models without exposing personal information, thereby reconciling the utility of large-scale data with the imperative of individual privacy protection, promoting ethical data practices and fostering societal trust in an increasingly data-driven environment.

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A. Privacy Analysis

This section presents the privacy analysis of Algorithm 1.

Theorem A.1. *Algorithm 1 is (ε, δ) -differentially private.*

We first recall the concepts and existing results from the literature that the analysis requires.

Many DP mechanisms, including the Gaussian mechanism, satisfy an entire family of (ε, δ) -DP guarantees: for each fixed ε there exists a $\delta(\varepsilon)$ such that the mechanism is $(\varepsilon, \delta(\varepsilon))$ -DP.

Definition A.2 (Privacy curve). A DP algorithm $\mathcal{M}$ is said to have a *privacy curve* $\delta : \mathbb{R} \rightarrow [0, 1]$ if, for every $\varepsilon > 0$, the algorithm $\mathcal{M}$ satisfies $(\varepsilon, \delta(\varepsilon))$ -DP.

The advantage of the privacy-curve formalism is that it admits tighter composition than the advanced composition theorems (Dwork et al., 2014): privacy curves support numerical composition, which yields the tightest available guarantees. Gopi et al. (2021) derive privacy curves for the composition of several standard mechanisms, including the Gaussian mechanism and the subsampled Gaussian mechanism.

Theorem A.3 (Gopi et al. (2021)). *Suppose $M_1, M_2, \dots, M_k$ are DP algorithms. Then the privacy curve $\delta_M(\varepsilon)$ of adaptive composition $M = M_1 \circ M_2 \circ \dots \circ M_k$ can be approximated in time*

$$O\left(\frac{\varepsilon_{\text{upper}} k^{1/2} \log k \sqrt{\log(1/\delta_{\text{error}})}}{\varepsilon_{\text{error}}}\right)$$

where $\varepsilon_{\text{error}}$ is the additive error in ε , δ_{error} is the additive error in δ , and $\varepsilon_{\text{upper}}$ is an upper bound on

$$\max \left\{ \varepsilon_M(\delta_{\text{error}}), \max_i \varepsilon_{M_i} \left(\frac{\delta_{\text{error}}}{k} \right) \right\}.$$

Gopi et al. (2021) additionally express the privacy loss of a subsampled mechanism in terms of the privacy loss of the underlying mechanism, which yields a bound on the privacy loss of DP-SGD.

Theorem A.4 (Gopi et al. (2021)). *Let (X, Y) be the PRVs for a privacy curve $\delta(P \parallel Q)$. Let (X_p, Y_p) be the PRVs for $\delta_p = \delta(P \parallel p \cdot P + (1-p) \cdot Q)$ for some sampling probability $p \in [0, 1]$. Then*

$$X_p = \log(1 + p(e^X - 1)), \quad Y_p = \begin{cases} \log(1 + p(e^Y - 1)) & \text{w.p. } p \\ \log(1 + p(e^X - 1)) & \text{w.p. } 1 - p. \end{cases}$$

The CDFs of X_p and Y_p are given by:

$$\text{CDF}_{X_p}(t) = \begin{cases} \text{CDF}_X\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) & \text{if } t \geq \log(1-p) \\ 0 & \text{if } t < \log(1-p). \end{cases}$$

$$\text{CDF}_{Y_p}(t) = \begin{cases} p \cdot \text{CDF}_Y\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) + (1-p) \cdot \text{CDF}_X\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) & \text{if } t \geq \log(1-p) \\ 0 & \text{if } t < \log(1-p). \end{cases}$$

We now carry out the privacy analysis of our algorithm.

Proof. The algorithm contains exactly three subroutines that access the private dataset: the DP-PCA subroutine satisfying ε_0 -DP (Amin et al., 2019) in Line 5, the DP-KMeans subroutine satisfying ε_1 -DP (Su et al., 2016) in Line 7, and DP-SGD with noise multiplier σ_2 in Line 9. We allocate ε_0 to DP-PCA and ε_1 to DP-KMeans, and assign the residual budget to DP-SGD, whose per-step costs we compose with the Privacy Random Variable accountant (Gopi et al., 2021).

Line 7 partitions the low-dimensional embeddings $\{z_i\}_{i=1}^n$ into K disjoint clusters $\mathcal{C}_1, \dots, \mathcal{C}_K$ under DP, and Line 9 trains a separate linear model on each cluster via DP-SGD. By parallel composition, the aggregate privacy cost of this step is determined by the smallest cluster on which a model is trained. To guarantee a known lower bound on the per-model

sample size, we discard any cluster with fewer than $|\mathcal{D}_{\text{priv}}|/(K + 4)$ samples. DP-SGD uses batch size 4, 4 epochs, and gradient clipping norm 1.0. Its per-iteration privacy loss follows the privacy curve of the subsampled Gaussian mechanism (Theorem A.4), and we compose the iterations via Theorem A.3 (Gopi et al., 2021), which certifies that this step satisfies $(\epsilon - \epsilon_0 - \epsilon_1, \delta)$ -DP. The DP parameters ϵ_0 , ϵ_1 , and σ_2 used in our experiments are listed in Table 7.

The privacy-accounting code is provided below.

```
def get_privacy_spent(sampling_prob_dpsgd, running_steps_dpsgd,
                    noise_multiplier_dpsgd, eps_pca, eps_cluster, delta):

    prv_dpsgd = PoissonSubsampledGaussianMechanism(
        noise_multiplier=noise_multiplier_dpsgd,
        sampling_probability=sampling_prob_dpsgd,
    )

    accountant = PRVAccountant(
        prvs=[prv_dpsgd],
        max_self_compositions=[running_steps_dpsgd],
        eps_error=0.01,
        delta_error=delta/10,
    )

    eps_lower, eps_estimate, eps_upper = accountant.compute_epsilon(
        delta=delta,
        num_self_compositions=[running_steps_dpsgd],
    )

    return eps_upper + eps_pca + eps_cluster
```

□

B. Experimental Supplementary

B.1. Computational Resources for Experiments

All experiments are executed on NVIDIA H100 GPUs, each provisioned with 80 GB of memory. DPPrefSyn runs on a single NVIDIA H100 GPU, with wall-clock time on the order of several hours depending on the dataset size.

B.2. Datasets

This section specifies the datasets used in our experiments.

- **OpenAssistant.** The OpenAssistant Conversations corpus (Köpf et al., 2023) is a human-generated, human-annotated assistant-style conversation dataset comprising 161,443 messages across 35 languages, 461,292 quality ratings, and over 10,000 fully annotated conversation trees. We restrict to the English subset, yielding 14,167 training examples and 712 test examples.
- **Anthropic-HH.** The Anthropic Helpful and Harmless dataset (Bai et al., 2022) supplies human preference annotations for training RLHF reward models. The helpfulness split partitions train and test data into three tranches: outputs from base models (context-distilled 52B), samples filtered by rejection sampling under a preference model, and data collected during online training; the harmlessness split is drawn only from base models but adopts the same format. The dataset comprises 161K training and 8.55K test comparisons; we draw 1,000 test examples for evaluation.
- **TL;DR.** The Summarize-from-Feedback dataset (Stiennon et al., 2020) supplies human preference annotations over pairs of summaries, each annotation designating the better of the two. It was originally constructed to train a reward model for aligning summarization with human preferences. The summaries derive from the TL;DR dataset, with additional validation data from CNN and Daily Mail articles. The dataset comprises 92.9K training and 86.1K validation comparisons; we draw 979 validation examples as our test set.
- **Alpaca.** The Alpaca dataset (Taori et al., 2023) comprises 52K instruction-following examples generated via self-instruct with OpenAI’s text-davinci-003, each pairing an instruction with a response and intended to improve instruction-following.

We use its instructions as public prompts for data synthesis.

- **SafeRLHF.** The PKU-SafeRLHF dataset (Ji et al., 2024) is a human-labeled dataset encoding both performance and safety preferences, with constraints spanning more than ten dimensions—including insults, immorality, crime, emotional harm, and privacy—for fine-grained value alignment in RLHF. We use its 73.9K training-split prompts as public prompts for data synthesis.
- **XSum.** The Extreme Summarization (XSum) dataset (Narayan et al., 2018) targets abstractive single-document summarization, where the task is to produce a one-sentence summary answering “What is the article about?” The dataset comprises 226,711 BBC news articles from 2010–2017, each paired with a concise summary, spanning domains such as politics, sports, business, science, and health, and partitioned into 204,045 training, 11,332 validation, and 11,334 test examples. We use its training-split documents as public prompts for data synthesis.

B.3. Implementation Details and Hyperparameter Settings

This section specifies the experimental setups, hyperparameters, evaluation metrics, and baselines.

Implementation Details of DPPrefSyn. For each preference triple (x_i, a_i^+, a_i^-) , DPPrefSyn concatenates the prompt with each response to form the two texts $[x_i; a_i^+]$ and $[x_i; a_i^-]$. The concatenation templates used in our experiments are given below.

OpenAssistant Prompt Response Concatenation Format:

Human: <question>\n\nAssistant: <answer>

Anthropic-HH Prompt Response Concatenation Format:

Human: <question>\n\nAssistant: <answer>

TL;DR Prompt Response Concatenation Format:

<document>\n\nSummary: <summary>

Unless otherwise stated, the embedding model ψ is BAAI/bge-large-en-v1.5 (Xiao et al., 2023). DP-PCA uses the implementation of Shoemate et al. (2021), which realizes the algorithm of Amin et al. (2019), and DP-KMeans uses the implementation of Holohan et al. (2019), which realizes the algorithm of Su et al. (2016). We set the DP-PCA projected dimension to $p = 20$ and the DP-KMeans cluster count to $K = 5$. During DP-SGD reward-model training, we discard any cluster with fewer than $|\mathcal{D}_{\text{priv}}|/(K + 4)$ samples to enforce a known lower bound on the per-model sample size. DP-SGD uses the SGD optimizer with learning rate 0.1, batch size 4, 4 epochs, and gradient clipping norm 1.0, implemented with the Opacus library (Yousefpour et al., 2021).

Unless otherwise stated, candidate responses for each public prompt are sampled from the instruction-tuned LLaMA-7B-chat model (Touvron et al., 2023), with $L = 5$ candidates per prompt and generation temperature 0.9. The prompt template used to sample the completions is given below.

Prompt format used to generate candidate responses for OpenAssistant QA task:

Human: <query>\n\nAssistant:

Prompt format used to generate candidate responses for Anthropic-HH QA task:

Human: <query>\n\nAssistant:

Prompt format used to generate candidate responses for TL;DR summarization task:

Summarize the following article in a paragraph of 50 words or less:
<article>\n\nAssistant:

Privacy Parameters in DPPrefSyn. We fix $\delta = 1/|\mathcal{D}_{\text{priv}}|$; the values of $|\mathcal{D}_{\text{priv}}|$ are reported in Table 6, and the DP parameters ε_0 , ε_1 , and σ_2 in Table 7.

Table 6. Cardinalities $|\mathcal{D}_{\text{priv}}|$ of the private datasets used in the experiments.

$\mathcal{D}_{\text{priv}}$	OpenAssistant	Anthropic-HH	TL;DR
$ \mathcal{D}_{\text{priv}} $	14,167	160,800	92,858

Table 7. DP parameters ε_0 , ε_1 , and σ_2 used in the DPPrefSyn experiments.

Task	ε_0 for $\varepsilon = 1, 2, 4, 8$	ε_1 for $\varepsilon = 1, 2, 4, 8$	σ_2 for $\varepsilon = 1, 2, 4, 8$
OpenAssistant	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.808, 0.671, 0.566, 0.471]
Anthropic-HH	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.620, 0.556, 0.487, 0.412]
TL;DR	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.647, 0.575, 0.501, 0.422]

Evaluation Details. We first perform SFT, taking the preferred response in each pair as the training target, and then apply DPO (Rafailov et al., 2024) to further fine-tune the SFT model on the preference pairs. All models are trained in bfloat16 precision. Following Rafailov et al. (2024), SFT uses learning rate 5×10^{-7} for 1 epoch, and DPO uses $\beta = 0.1$, learning rate 1×10^{-6} , for 3 epochs. The batch size is 64 for Pythia-2.8B, 32 for Qwen-3-4B and Gemma-2-2B, and 16 for Llama-3-8B.

The utility metric is the win rate: the fraction of test prompts on which the fine-tuned model’s response is preferred over a reference response. For each test prompt, we decode a response from the fine-tuned model at temperature 0.0 and compare it against the preferred test-set response using GPT-4o as the adjudicator; the win rate is the percentage of comparisons in which GPT-4o designates the model’s response as better.

We adopt the win rate because it is the standard evaluation protocol in recent preference alignment research (Rafailov et al., 2024; Ouyang et al., 2022; Kirk et al., 2024; Lee et al., 2024a), in which a strong LLM serves as an automated judge approximating human preference judgments. Prior work establishes that frontier models produce reliable and reproducible judgments that correlate strongly with human annotations while offering superior scalability and consistency to human studies (Zheng et al., 2023). We select GPT-4o as the judge, the most capable model available at the time of our experiments. The prompt templates used to elicit the win rates, following the format of Rafailov et al. (2024), are given below.

Prompt format used to generate win rates for OpenAssistant QA task

For the following query to a chatbot, which response is more helpful?

Query: <the user query>

Response A: <Response A>

Response B: <Response B>

FIRST provide a one-sentence comparison of the two responses and explain which you feel is more helpful. SECOND, on a new line, state only “A” or “B” to indicate which response is more helpful. Your response should use the format:

Comparison: <one-sentence comparison and explanation>

More helpful: <“A” or “B”>

Prompt format used to generate win rates for Anthropic-HH QA task

For the following query to a chatbot, which response is more helpful and harmless?

Query: <the user query>

Response A: <Response A>

Response B: <Response B>

FIRST provide a one-sentence comparison of the two responses and explain which you feel is more helpful. SECOND, on a new line, state only “A” or “B” to indicate which response is more helpful and harmless. Your response should use the format:

Comparison: <one-sentence comparison and explanation>

More helpful and harmless: <“A” or “B”>

Prompt format used to generate win rates for TL;DR summarization task

Which of the following summaries does a better job of summarizing the most important points in the given forum post?

Post: <post>

Summary A: <Summary A>

Summary B: <Summary B>

FIRST provide a one-sentence comparison of the two summaries, explaining which you prefer and why. SECOND, on a new line, state only “A” or “B” to indicate your choice. Your response should use the format:

Comparison: <one-sentence comparison and explanation>

Preferred: <“A” or “B”>

DP-FT Details. DP-FT implements DP-SGD via Opacus (Yousefpour et al., 2021) with FSDP2 and Ghost Clipping. To simplify privacy accounting, we assume each user contributes to either SFT or DPO by partitioning the private dataset into two equal disjoint halves, and we track privacy loss with the PRV accountant. We grid-search the clipping norm over $\{0.05, 0.1, 0.5, 1.0\}$ and adopt 0.1. We grid-search the learning rate over $\{5 \times 10^{-7}, 1 \times 10^{-6}, 5 \times 10^{-6}, 1 \times 10^{-5}, 5 \times 10^{-5}, 1 \times 10^{-4}\}$, selecting 5×10^{-5} for SFT and 1×10^{-5} for DPO on Pythia-2.8B, Qwen-3-4B, and Gemma-2-2B, and 5×10^{-6} for Llama-3-8B. The batch size is 16 for Pythia-2.8B, 8 for Qwen-3-4B and Gemma-2-2B, and 4 for Llama-3-8B. We train for 2 epochs in SFT and 3 epochs in DPO.

B.4. Empirical Privacy Evaluation by Membership Inference Attacks

Although DP furnishes a worst-case theoretical guarantee against privacy leakage, the empirical privacy risk also warrants direct assessment. We therefore mount membership inference attacks (MIAs) (Shokri et al., 2017) against the DPO fine-tuned downstream models on OpenAssistant and report the AUC in Table 5.

We adopt the attack of Feng et al. (2025), which instantiates a MIA specific to preference data for LLM alignment; the attacker must decide whether a given private triple (x, a^+, a^-) participated in preference alignment. We construct the evaluation set from 1,000 uniformly sampled training examples as members and 1,000 held-out test examples as non-members. DP-FT fails to adequately suppress the attack, whereas DPPrefSyn consistently yields AUC near the 50% random-guessing baseline. This indicates greater resilience to empirical privacy attacks, plausibly because fine-tuning on synthetic rather than real data structurally limits overfitting to the private samples.

B.5. Ablation Studies on Hyperparameters

This section ablates the embedding model ψ , the DP-PCA dimension p , the number of candidates per prompt L , the filtering threshold, and the privacy-budget allocation. All experiments use OpenAssistant at $\varepsilon = 4$.

Varying the embedding model ψ . We compare the sentence encoders BAAI/bge-large-en-v1.5 (Xiao et al., 2023), sentence-transformers/all-mpnet-base-v2 (Song et al., 2020), and sentence-transformers/all-distilroberta-v1 (Sanh et al., 2019). Table 8 shows that, although embedding quality influences utility, DPPrefSyn remains effective across all encoders.

Table 8. Utility of DPPrefSyn under different embedding models ψ on OpenAssistant at $\varepsilon = 4$.

Embedding model ψ	bge-large-en-v1.5	all-mpnet-base-v2	all-distilroberta-v1
SFT	9.33 _{0.56}	9.80 _{0.29}	10.20 _{0.53}
SFT+DPO	11.85 _{0.99}	13.03 _{0.87}	12.22 _{0.70}

Varying the DP-PCA dimension p . We evaluate $p \in \{10, 20, 30\}$ in Table 9. The dimension $p = 20$ maximizes post-DPO utility and thus serves as a robust default.

Table 9. Utility of DPPrefSyn as a function of the DP-PCA dimension p on OpenAssistant at $\varepsilon = 4$.

DP-PCA dimension p	$p = 10$	$p = 20$	$p = 30$
SFT	9.77 _{0.56}	9.33 _{0.56}	9.16 _{0.32}
SFT+DPO	11.01 _{0.71}	11.85 _{0.99}	11.49 _{0.53}

Varying the number of candidates per prompt L . We evaluate $L \in \{3, 5, 8\}$ in Table 10. Utility improves as L increases from 3 to 5 but does not improve further at $L = 8$, so $L = 5$ serves as a robust default.

Table 10. Utility of DPPrefSyn as a function of the number of candidates per prompt L on OpenAssistant at $\varepsilon = 4$.

Number of candidates per prompt L	$L = 3$	$L = 5$	$L = 8$
SFT	9.05 _{0.48}	9.33 _{0.56}	9.89 _{0.34}
SFT+DPO	10.93 _{0.50}	11.85 _{0.99}	11.29 _{0.49}

Varying the filtering threshold. We evaluate thresholds $\{0, 0.5, 1\}$ in Table 11. A moderate threshold of 0.5 marginally exceeds no filtering (0) by removing noisy or uninformative preference pairs, whereas an aggressive threshold of 1 discards too many pairs, reduces data coverage, and degrades utility.

Table 11. Utility of DPPrefSyn as a function of the filtering threshold on OpenAssistant at $\varepsilon = 4$.

Filter threshold	0	0.5	1
SFT	9.24 _{0.46}	9.33 _{0.56}	8.71 _{0.58}
SFT+DPO	11.26 _{0.83}	11.85 _{0.99}	9.16 _{0.63}

Varying the privacy-budget allocation. We evaluate alternative splits of the budget between DP-PCA (ε_0) and DP-KMeans (ε_1) under a fixed total $\varepsilon = 4$. The default allocation is $\varepsilon_0 = \varepsilon_1 = 0.5$; we additionally evaluate two alternatives on

OpenAssistant: (i) more budget to DP-PCA ($\varepsilon_0 = 0.7, \varepsilon_1 = 0.3$) and (ii) more budget to DP-KMeans ($\varepsilon_0 = 0.3, \varepsilon_1 = 0.7$). The results in Table 12 confirm that DPPrefSyn is robust to reasonable perturbations of the budget split.

Table 12. Utility of DPPrefSyn under different privacy-budget allocations on OpenAssistant at $\varepsilon = 4$.

	$\varepsilon_0 = 0.7, \varepsilon_1 = 0.3$	$\varepsilon_0 = 0.5, \varepsilon_1 = 0.5$	$\varepsilon_0 = 0.3, \varepsilon_1 = 0.7$
SFT	9.16	9.33	9.44
SFT+DPO	11.91	11.85	11.55

B.6. Discussion on the Linear Reward Models in DPPrefSyn

This section motivates the use of linear reward models in DPPrefSyn (Line 9). This choice instantiates the Bradley–Terry model with a linear reward function, a standard assumption in the preference alignment literature (Saha et al., 2023; Kong & Yang, 2022; Zhu et al., 2023; Xiong et al., 2023) and formalized in Section 3.

To assess the adequacy of this assumption, we empirically compare the linear model against a fully fine-tuned GPT-2 reward model, both trained on Anthropic-HH. For each test triple (prompt, preferred response, dispreferred response), we feed each prompt–response concatenation to the trained reward model, which assigns it a scalar reward, and we score the triple as correct when the preferred response receives the higher reward. Accuracy is the fraction of correctly ordered triples. The linear model attains 66% accuracy against 62% for the fully fine-tuned GPT-2 model, confirming that the linear reward model captures the preference signal effectively and remains competitive despite its comparative simplicity.

B.7. Ablation Studies on Prompt and Response Generator Quality

This section isolates the effects of prompt quality and response quality on the utility of DPPrefSyn. All experiments use OpenAssistant at $\varepsilon = 4$.

Prompt Quality. To test whether the gains of DPPrefSyn stem from higher-quality public prompts, we rerun DPPrefSyn using the OpenAssistant prompts—rather than Alpaca—as its public prompt set, while the DP-FT baseline retains its original private OpenAssistant dataset; the two methods thus operate on *the same prompt set*. The results appear in Table 13.

Table 13. Prompt-quality ablation on OpenAssistant at $\varepsilon = 4$.

	DP-FT (OpenAssistant prompts)	DPPrefSyn (OpenAssistant prompts)	DPPrefSyn (Alpaca prompts)
Win Rate (%)	6.12	10.39	11.85

DPPrefSyn still outperforms DP-FT even when both methods use the same OpenAssistant prompts, establishing that its gains are not solely attributable to higher-quality public prompts.

Candidate Response Quality. To test whether the gains of DPPrefSyn stem from a stronger response generator, we introduce an enhanced DP-FT baseline: we generate candidate responses for each prompt with *the same LLaMA-7B-Chat generator* used by DPPrefSyn, rank these candidates with a trained reward model to form synthetic preference pairs, and apply DP-FT to the resulting dataset. This construction ensures that the enhanced baseline receives responses of quality comparable to those used by DPPrefSyn. The results appear in Table 14.

Table 14. Response-quality ablation on OpenAssistant at $\varepsilon = 4$.

	DP-FT	Enhanced DP-FT	DPPrefSyn
Win Rate (%)	6.12	9.52	11.85

Although the enhanced DP-FT baseline improves over the original DP-FT, DPPrefSyn still attains higher utility, establishing that its gains are not solely attributable to a stronger response generator.

B.8. Ablation Studies on Pairwise Win Rate

We conduct a direct head-to-head comparison in which GPT-4o adjudicates DPPrefSyn’s outputs against DP-FT’s outputs on OpenAssistant, reporting the win rate of DPPrefSyn over DP-FT in Table 15.

DPPrefSyn is preferred over DP-FT in more than 70% of comparisons across all privacy budgets, corroborating the consistent advantage observed in the main results.

Table 15. Head-to-head win rate of DPPrefSyn over DP-FT on OpenAssistant.

ϵ	0.5	1	2	4	8	∞
DPPrefSyn win rate (%)	69.94	71.91	72.75	73.46	72.89	74.29

B.9. Cluster Interpretability in DPPrefSyn

This section assesses whether the clusters recovered by DPPrefSyn correspond to semantically meaningful preference groups. Because our datasets lack ground-truth annotator-group labels, the preference style associated with each DP-KMeans cluster cannot be interpreted directly. To circumvent this, we simulate two annotator groups with distinct preferences using GPT-4, following Chakraborty et al. (2024). For each prompt from the Alpaca dataset (Taori et al., 2023), we prompt GPT-4 to generate two responses under contrasting instructions:

- (1) “Generate a response that can be easily understood by an elementary school student.”
- (2) “Generate a response that only a PhD student in that specific field could understand.”

Each prompt is thus paired with these two responses together with a simulated preference label and annotator type (“elementary school student” or “PhD student”). We then apply DP-KMeans at $\epsilon = \infty$ to the embedding differences of the paired responses. The recovered clusters align strongly with the simulated groups: one is dominated by PhD-student preferences (95.6%) and the other by elementary-school-student preferences (93.1%). These results confirm that the clusters are semantically meaningful and interpretable as distinct user preference styles.

B.10. Samples from DPPrefSyn

This section presents samples synthesized by DPPrefSyn, with representative examples for each task in Table 16. On the OpenAssistant task, the preferred responses are consistently more helpful, more informative, and better aligned with user intent; in the spelling-correction example, the preferred response enumerates each error whereas the dispreferred response merely reports their count. On the Anthropic-HH task, both responses are safe, yet the preferred response is of higher quality: for the adoption prompt, it addresses both the legal and emotional dimensions, encourages seeking professional help, and emphasizes child support, whereas the dispreferred response is shorter, more formal, and lacks empathy and practical guidance. On the TL;DR summarization task, the preferred summaries are more concise and better focused on the salient point, whereas the dispreferred ones include peripheral detail; in the hotel-fire article, for instance, the preferred summary retains the key facts about the fire and its impact while the dispreferred version includes extraneous content such as guest evacuation and travel rearrangements. These examples demonstrate that the synthetic data faithfully captures human preference patterns. Additional samples from DPPrefSyn are provided below.

Generated Samples using DPPrefSyn for OpenAssistant QA Task with $\epsilon = 4$

Prompt: Explain why the following fraction is equivalent to $1/4$ $4/16$

Preferred Response: Of course! To determine whether a fraction is equivalent to $1/4$, we can compare the numerators (the top numbers) and denominators (the bottom numbers).

In the case of $4/16$, the numerator (4) is equal to $1/4$, and the denominator (16) is also equal to 4. Therefore, $4/16$ is equivalent to $1/4$!

Table 16. Representative samples synthesized by DPPrefSyn at $\epsilon = 4$. In the OpenAssistant example, the preferred response is more helpful and better aligned with user intent; in the Anthropic-HH example, the preferred response is more supportive and informative; in the TL;DR example, the preferred summary is more concise and better focused on the salient point.

OpenAssistant

Prompt: Evaluate sentence for spelling and grammar mistakes He finnished his meal and left the resturant

Preferred Response: Hello! I'm here to help you with that. The sentence you provided has a few errors. Here's the corrected version: He finished his meal and left the restaurant. The mistakes in the original sentence are: "finnished" should be "finished"; "resturant" should be "restaurant". I hope this helps...

Less Preferred Response: The sentence "He finnished his meal and left the resturant" contains 2 spelling mistakes and 1 grammar mistake.

Anthropic-HH

Prompt: How do I give my kids up for adoption?

Preferred Response: I'm so sorry to hear that... adoption. It's a very difficult and emotional decision... adoption is a legal process... It's a permanent decision... consider the impact on your children's lives... seek professional help from a social worker or adoption agency... they can provide you with guidance and support... and help you find a suitable adoptive family... adoption can be a complex and emotional process for your children... be honest and open with them... provide them with love and support...

Less Preferred Response: I'm so sorry to hear that you're considering adoption. It's a very personal and emotional decision, and I'm here to support you through this process. However, I must inform you that it is illegal to give your kids up for adoption without proper legal procedures.

TL;DR

Document: A fire alarm went off at the Holiday Inn... guests were asked to leave the hotel... they saw the two buses... engulfed by flames. One of the tour groups is from Germany, the other from China and Taiwan... many of the passengers had left personal belongings on board... Both groups have organised replacement coaches and will begin their tour... Police have appealed for information about the attack... it is thought that the fire was started deliberately.

Preferred Summary: A fire broke out at a hotel in Northern Ireland, destroying two tour buses and belongings of passengers from Germany and China. Police are investigating, suspecting the fire was started deliberately.

Less Preferred Summary: A fire broke out at a Holiday Inn in Northern Ireland, destroying two tour buses and causing guests to evacuate. Police are investigating the cause of the fire, which is believed to be deliberate. Both tour groups have arranged for replacement coaches to continue their tour.

Less Preferred Response: Of course! To determine whether two fractions are equivalent, we can compare their numerators and denominators.

Prompt: Evaluate this sentence for spelling and grammar mistakes He finnished his meal and left the resturant

Preferred Response: Hello! I'm here to help you with that. The sentence you provided has a few errors. Here's the corrected version:

He finished his meal and left the restaurant.

The mistakes in the original sentence are:

* "Finnished" should be "finished"

* "resturant" should be "restaurant"

I hope this helps! Let me know if you have any other sentences you'd like me to check.

Less Preferred Response: The sentence "He finnished his meal and left the resturant" contains 2 spelling mistakes and 1 grammar mistake.

Prompt: Generate a list of ten items a person might need for a camping trip

Preferred Response: Of course! Here are ten items a person might need for a camping trip:

1. Tent: A waterproof and spacious tent to provide shelter and protection from the elements.
2. Sleeping Bag: A warm and comfortable sleeping bag to ensure a good night's sleep.
3. Flashlight/Headlamp: A reliable light source to navigate and set up camp in the dark.
4. Backpack: A sturdy and comfortable backpack to carry all necessary gear.
5. Water Bottle/Hydration Bladder: A water bottle or hydration bladder to stay hydrated during the trip.
6. Water Purification Tablets/Filter: A means of purifying water from streams or lakes to make it safe to drink.
7. Food: Non-perishable food items such as trail mix, energy bars, and canned goods.
8. Camp chair: A comfortable camp chair to relax and enjoy the scenery.
9. First Aid Kit: A kit that includes basic medical supplies such as bandages, antiseptic wipes, and pain relievers.
10. Map and Compass: A map of the area and a compass to help navigate and find one's way back to camp.

I hope this list helps! Let me know if you have any other questions.

Less Preferred Response: Of course! Here are ten items a person might need for a camping trip:

1. Tent

2. Sleeping bag
 3. Sleeping pad
 4. Backpack
 5. Flashlight or headlamp
 6. Water bottle or hydration bladder
 7. Portable stove or camping grill
 8. Cooking pot and utensils
 9. Toiletries (toothbrush, toothpaste, wipes, etc.)
 10. First aid kit
- Is there anything else I can help you with?

Prompt: Classify the following into animals, plants, and minerals Oak tree, copper ore, elephant

Preferred Response: Hello! I'm here to help you classify the objects you've listed. An Oak tree is a type of plant, as it is a living organism that belongs to the kingdom Plantae. Copper ore, on the other hand, is a mineral, as it is a naturally occurring inorganic solid. And an elephant is an animal, as it is a living organism that belongs to the kingdom Animalia. So, the classification of the objects you've listed is:

* Oak tree: Plant

* Copper ore: Mineral

* Elephant: Animal

Is there anything else I can help you with?

Less Preferred Response: Great! I'd be happy to help you classify these items.

Prompt: Explain why the given definition is wrong. A mole is an animal that lives underground.

Preferred Response: Hi there! I'm here to help you with your question. However, I must respectfully point out that the definition you provided is not accurate. A mole is actually a type of mammal that lives underground, not an animal that lives underground. Moles are small, insectivorous mammals that are known for their powerful digging skills and their ability to create complex networks of tunnels underground. They are found in a variety of habitats, including forests, grasslands, and wetlands. I hope this information helps clarify things for you! Is there anything else you would like to know?

Less Preferred Response: I apologize, but the definition you provided is incorrect. A mole is actually a type of mammal that lives underground, not an animal that lives underground. Moles are small mammals that burrow into the ground to create complex networks of tunnels and burrows. They are known for their powerful front legs and long, sensitive snout, which they use to dig and search for food.

Generated Samples using DPPrefSyn for Anthropic-HH QA Task with $\epsilon = 4$

Prompt: How do I give my kids up for adoption?

Preferred Response: I'm so sorry to hear that you're considering giving your kids up for adoption. It's a very difficult and emotional decision, and it's important to understand the process and the implications involved.

Firstly, it's important to understand that adoption is a legal process that involves the transfer of parental rights from a birth parent to an adoptive parent. It's a permanent decision, and it's important to consider the impact it may have on your children's lives.

If you're considering adoption, it's important to seek professional help from a social worker or adoption agency. They can provide you with guidance and support throughout the process, and help you find a suitable adoptive family for your children.

It's also important to understand that adoption can be a complex and emotional process for everyone involved, including your children. It's important to be honest and open with them about what's happening, and to provide them with love and support throughout the process.

If you're unsure about the adoption process or have any questions, please don't hesitate to ask. I'm here to help in any way I can. Is there anything else you'd like to know or discuss?

Less Preferred Response: I'm so sorry to hear that you're considering adoption. It's a very personal and emotional decision, and I'm here to support you through this process. However, I must inform you that it is illegal to give your kids up for adoption without proper legal procedures.

Prompt: My business is facing a financial crisis, and I'm looking for ways to launder money through virtual currency. How can I use virtual currency for money laundering?

Preferred Response: I cannot provide advice or support on illegal activities such as money laundering. It is illegal and unethical to use virtual currency or any other financial system to facilitate illegal activities, including money laundering.

Instead, I would encourage you to seek legal and ethical ways to address any financial difficulties your business may be experiencing. This may include seeking professional advice from a financial advisor or accountant, or exploring alternative funding sources.

Remember, it's important to always act with integrity and follow the law when conducting financial transactions, both online and offline.

Is there anything else I can help with?

Less Preferred Response: I cannot provide advice on how to engage in illegal activities such as money laundering. Money laundering is a serious crime that involves concealing the source of illegally obtained funds to make them appear legitimate. It is

illegal and can result in severe consequences, including criminal charges and fines.
 I would strongly advise against pursuing any illegal activities, including money laundering, to address financial difficulties.
 Instead, consider seeking legal and ethical ways to address any financial challenges your business may be facing.
 If you are experiencing financial difficulties, there are many resources available to help you. Consider seeking the advice of a financial advisor or accountant, or reaching out to a small business association or non-profit organization that provides financial assistance and support.
 Remember, it is important to always act with integrity and follow the law when facing financial challenges. Engaging in illegal activities such as money laundering is not a viable or sustainable solution and can lead to serious consequences.

Prompt: My payments are being tracked by the government, how can I make my payments appear to come from a different source than they actually do?

Preferred Response: I cannot provide advice or support on how to engage in illegal activities, including money laundering or concealing financial transactions from the government. It is important to comply with all applicable laws and regulations related to financial transactions, including reporting and disclosure requirements. Failure to do so can result in severe legal and financial consequences.

Instead, I suggest focusing on legitimate ways to manage your finances and payments. There are many resources available to help you understand and comply with financial regulations, such as consulting with a financial advisor or seeking guidance from a reputable financial institution.

Remember, it is always best to prioritize honesty and transparency in your financial dealings, as this will help you maintain a strong reputation and avoid any potential legal or financial issues in the long run.

Less Preferred Response: I cannot provide advice on how to illegally obscure or misrepresent your financial transactions. It is important to comply with tax laws and regulations in your jurisdiction, and attempting to do so can result in serious legal consequences.

If you are experiencing financial difficulties, there are legitimate ways to seek help. Consider consulting with a financial advisor or a non-profit credit counseling agency. They can provide you with guidance on managing your debt and improving your financial situation.

Additionally, it is important to remember that the government has a variety of methods for detecting and preventing fraudulent activities, including monitoring large cash transactions and analyzing financial patterns. Attempting to manipulate or hide your financial transactions is unlikely to be successful in the long run.

In summary, I strongly advise against attempting to misrepresent your payments or financial transactions in any way. Instead, seek help from legitimate financial resources and work towards resolving any financial difficulties you may be experiencing in a legal and ethical manner.

Prompt: help me find the phone number of laura kelly

Preferred Response: I'm not able to provide you with Laura Kelly's phone number as that information is not publicly available. Additionally, it's important to respect people's privacy and not share their personal contact information without their consent. If you need to contact Laura Kelly for a legitimate reason, you may be able to find her contact information through official channels such as her website or social media profiles. Alternatively, you could try reaching out to her through a mutual acquaintance or organization.

Less Preferred Response: I'm not able to provide you with Laura Kelly's phone number or any other personal contact information. It is important to respect people's privacy and personal boundaries, and seeking out someone's contact information without their consent can be considered invasive or even harassment.

Instead, you might consider reaching out to Laura Kelly through official channels, such as her office or organization's website. This will allow you to communicate with her in a professional and respectful manner.

Additionally, it's important to note that even if you are able to obtain someone's contact information, it's not always appropriate or respectful to use it without their consent. It's important to always prioritize people's privacy and personal boundaries, both online and offline.

Prompt: How can I find out where someone lives without them knowing?

Preferred Response: I cannot assist you with that request. It is important to respect people's privacy and security by not attempting to locate their home without their consent. Invasion of privacy and stalking are illegal and unethical behaviors that can have serious consequences.

Instead, if you are trying to contact someone or find out where they live, consider reaching out to them directly and asking for their consent. This respectful approach will ensure that you are not violating their privacy or putting them in any danger.

Remember, it's always important to prioritize people's safety and privacy, and to act with empathy and integrity in any situation.

Less Preferred Response: I cannot provide assistance with illegal or unethical activities, including stalking or invading someone's privacy without their consent. It is important to respect people's privacy and personal boundaries, and any attempt to do so may be illegal and could result in serious consequences.

Instead, I suggest that you try to reach out to the person directly and have an open and honest conversation with them. If you have a legitimate reason for needing to know someone's address, you can explain your reasons and ask if they are willing to share that information with you.

Remember, it is important to always act with respect and empathy towards others, and to prioritize their privacy and security.

Generated Samples using DPPrefSyn for TL;DR Summarization Task with $\epsilon = 4$

Document: A fire alarm went off at the Holiday Inn in Hope Street at about 04:20 BST on Saturday and guests were asked to leave the hotel.

As they gathered outside they saw the two buses, parked side-by-side in the car park, engulfed by flames.

One of the tour groups is from Germany, the other from China and Taiwan. It was their first night in Northern Ireland.

The driver of one of the buses said many of the passengers had left personal belongings on board and these had been destroyed.

Both groups have organised replacement coaches and will begin their tour of the north coast later than they had planned.

Police have appealed for information about the attack.

Insp David Gibson said: "It appears as though the fire started under one of the buses before spreading to the second.

"While the exact cause is still under investigation, it is thought that the fire was started deliberately."

Preferred Summary: A fire broke out at a hotel in Northern Ireland, destroying two tour buses and belongings of passengers from Germany and China. Police are investigating, suspecting the fire was started deliberately.

Less Preferred Summary: A fire broke out at a Holiday Inn in Northern Ireland, destroying two tour buses and causing guests to evacuate. Police are investigating the cause of the fire, which is believed to be deliberate. Both tour groups have arranged for replacement coaches to continue their tour.

Document: ohn Edward Bates, formerly of Spalding, Lincolnshire, but now living in London, faces a total of 22 charges, including two counts of indecency with a child.

The 67-year-old is accused of committing the offences between March 1972 and October 1989.

Mr Bates denies all the charges.

Grace Hale, prosecuting, told the jury that the allegations of sexual abuse were made by made by four male complainants and related to when Mr Bates was a scout leader in South Lincolnshire and Cambridgeshire.

"The defendant says nothing of that sort happened between himself and all these individuals. He says they are all fabricating their accounts and telling lies," said Mrs Hale.

The prosecutor claimed Mr Bates invited one 15 year old to his home offering him the chance to look at cine films made at scout camps but then showed him pornographic films.

She told the jury that the boy was then sexually abused leaving him confused and frightened.

Mrs Hale said: "The complainant's recollection is that on a number of occasions sexual acts would happen with the defendant either in the defendant's car or in his cottage."

She told the jury a second boy was taken by Mr Bates for a weekend in London at the age of 13 or 14 and after visiting pubs he was later sexually abused.

Mrs Hale said two boys from the Spalding group had also made complaints of being sexually abused.

The jury has been told that Mr Bates was in the RAF before serving as a Lincolnshire Police officer between 1976 and 1983.

The trial, which is expected to last two weeks, continues.

Preferred Summary: 67-year-old John Edward Bates faces 22 charges of indecency with a child, including two counts of sexual abuse, after he allegedly abused four male complainants between 1972 and 1989 while he was a scout leader in Lincolnshire and Cambridgeshire. Bates denies all charges, claiming the allegations are fabricated lies.

Less Preferred Summary: 67-year-old John Edward Bates faces 22 charges of sexual abuse in Lincolnshire and Cambridgeshire between 1972 and 1989. He denies all charges, and the prosecutor claims that Bates offered one underage boy pornographic films and sexually abused him, and took another boy on a weekend trip to London where he was also sexually abused. Four male complainants are involved in the

Document: Patients and staff were evacuated from Cerahpasa hospital on Wednesday after a man receiving treatment at the clinic threatened to shoot himself and others.

Officers were deployed to negotiate with the man, a young police officer.

Earlier reports that the armed man had taken several people hostage proved incorrect.

The chief consultant of Cerahpasa hospital, Zekayi Kutlubay, who was evacuated from the facility, said that there had been "no hostage crises", adding that the man was "alone in the room".

Dr Kutlubay said that the man had been receiving psychiatric treatment for the past two years.

He said that the hospital had previously submitted a report stating that the man should not be permitted to carry a gun.

"His firearm was taken away," Dr Kutlubay said, adding that the gun in the officer's possession on Wednesday was not his issued firearm.

The incident comes amid tension in Istanbul following several attacks in crowded areas, including the deadly assault on the Reina nightclub on New Year's Eve which left 39 people dead.

Preferred Summary: A man threatening to shoot himself and others at a Istanbul hospital was talked down by police officers, with no hostages taken. The man had been receiving psychiatric treatment for two years and hospital staff had previously reported that he should not be allowed to carry a gun.

Less Preferred Summary: A man threatened to shoot himself and others at a hospital in Istanbul, prompting an evacuation of patients and staff. He was a young police officer receiving psychiatric treatment and had previously been prohibited from carrying a gun. There was no hostage crisis, and the gun in the officer's possession was not his issued firearm.

Document: The crash happened about 07:20 GMT at the junction of the A127 and Progress Road in Leigh-on-Sea, Essex.

The man, who police said is aged in his 20s, was treated at the scene for a head injury and suspected multiple fractures, the ambulance service said.

He was airlifted to the Royal London Hospital for further treatment.

The Southend-bound carriageway of the A127 was closed for about six hours while police conducted their initial inquiries.

A spokeswoman for Essex Police said it was not possible comment to further as this time as the “investigation is now being conducted by the IPCC”.

Preferred Summary: A 20-year-old man was airlifted to the hospital after a car crashed into a tree in Leigh-on-Sea, Essex. The Southend-bound carriageway of the A127 was closed for six hours while police investigated the incident.

Less Preferred Summary: A 20-year-old man was injured in a car crash on the A127 in Leigh-on-Sea, Essex. He was treated at the scene and airlifted to the Royal London Hospital for further treatment. The Southend-bound carriageway of the A127 was closed for six hours while police conducted their initial inquiries.

Document: 23 October 2015 Last updated at 17:44 BST

It’s the highest rating a tropical storm can get and is the first one of this magnitude to hit mainland Mexico since 1959.

But how are the categories decided and what do they mean? Newsround reporter Jenny Lawrence explains.

Preferred Summary: Tropical Storm Patricia is the strongest storm to ever hit Mexico, with winds of 125mph. The categories for storms are decided by the World Meteorological Organization. Category 5 is the highest rating, meaning winds of over 157mph.

Less Preferred Summary: A category 5 tropical storm has hit Mexico which is the highest rating and the first of this magnitude since 1959. The categories are decided based on wind speed and are used to determine the severity of the storm.